IOWA STATE UNIVERSITY

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AI-Powered Financial & Scholarship Assistant

Project Background

The Challenge
Navigating the complex landscape of scholarship applications and financial literacy is a significant hurdle for students in higher education. Students often face "information overload" when reviewing eligibility criteria, deadlines, and application requirements, leading to missed opportunities and increased stress.

The Solution
This project introduces a **24/7 Personal AI Assistant** designed to simplify academic and financial planning. By integrating a secure digital document library with advanced AI models, the system provides students with a streamlined way to manage their scholarship journey.

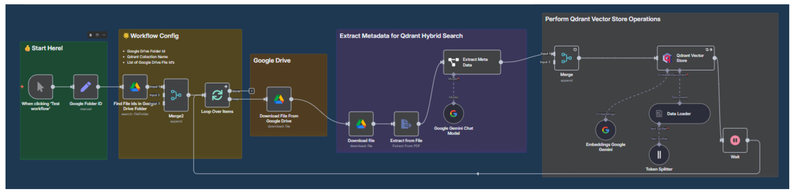
Key Features
- **Intelligent Document Retrieval**: Students can ask natural language questions about their stored documents and receive instant, context-aware answers with cited sources.
- **Automated Document Review**: The AI acts as a specialized advisor, evaluating uploaded resumes and essays against specific scholarship rubrics to provide actionable, real-time feedback.
- **24/7 Accessibility**: By automating manual document analysis, the system provides personalized advising at any time, empowering students to build stronger applications and improve their financial knowledge.

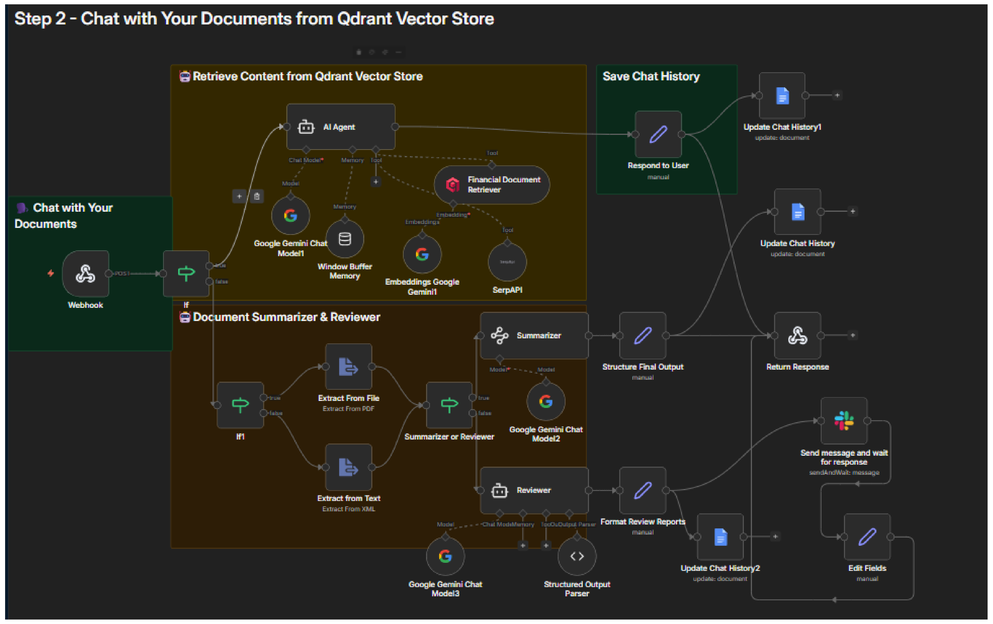
Methods

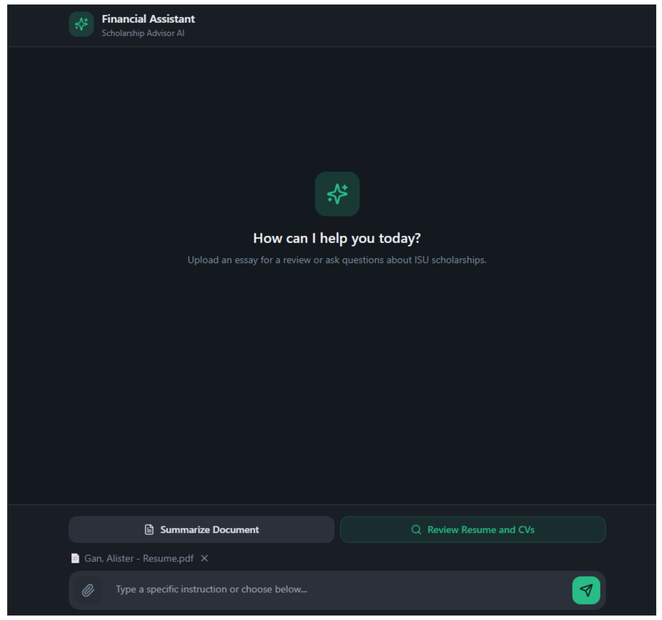
1. Document processing and storage
The workflow automatically gathers documents from a student's **Google Drive**, such as scholarship guides, financial aid policies, and personal notes. These files are processed by **Google Gemini AI** to extract important themes and stored in a digital library for instant access.

Path 1: Chat with your Documents Path (R.A.G)
The path is designed for deep research and knowledge retrieval.
- **What it does:** It acts like a personal librarian for your documents.
- **How it works**: When you ask a question, the system searches through a "Vector Store" (a digital memory bank) to find the exact pages or paragraphs that contain the answer. It then hands this information to the AI, which writes a custom response based only on the facts found in those documents.
- **Best for:** Answering questions like "What are the eligibility requirements in this policy document?" or "Summarize the key findings from these research papers".

Path 2: Reviewer & Summarizer Path
The path is designed for structured evaluation and feedback.
- **What it does:** It acts like a professional editor or coach.
- **How it works:** Instead of just searching for facts, this path breaks the document down into its core components (like structure, clarity, and tone). It runs the document through an AI "Reviewer" that assesses it against a specific rubric such as checking if a scholarship essay meets all program requirements and then generates a polished summary and actionable advice.
- **Best for:** Tasks like reviewing resumes and CVs and summarizing long reports into professional feedback.





Conclusion
- **Increased Accessibility**: Provides students with 24/7 access to personalized scholarship guidance and document review, removing the wait times associated with traditional advising.
- **Enhanced Application Quality**: Delivers consistent, objective, and actionable feedback on resumes and essays, helping students better align their drafts with scholarship rubrics and institutional values.
- **Reduced Information Overload**: Effectively manages and summarizes large volumes of financial data, allowing students to focus on high-impact application tasks rather than manual research.

Future Work
- Automated schedule to scan scholarship database and send email alerts to students before deadline.
- OneApp Connectivity by exploring API integrations to allow direct submission of AI-reviewed essays and resume in OneApp.